

Steel, Ships and Heavy Industries

Korea's Industrial Backbone in the Age of Transition

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Executive Summary

Korea's shipbuilding industry recorded its strongest financial performance in two decades in 2025. HD Korea Shipbuilding & Offshore Engineering (HD KSOE) — the holding company of Hyundai Heavy Industries, the world's largest shipbuilder — reported KRW 29.93 trillion (\$22.1B) revenue in 2025, an increase of 17.2% year-on-year, with operating profit of KRW 3.9 trillion (\$2.9B), up 172.3% from 2024. Samsung Heavy Industries and Hanwha Ocean (formerly DSME) similarly reported strong order books through 2028-2029, with backlogs sufficient to ensure full capacity utilisation at current pricing through mid-decade.

The recovery has been driven by a structural inflection: the global shipping fleet's transition to LNG-fuelled and, increasingly, ammonia and methanol dual-fuel vessels has created a product category where Korean shipbuilders hold decisive technology and production quality advantages over their Chinese competitors. Korean LNG carrier order share exceeds 90% globally — a dominance that persists even as Chinese builders have aggressively competed for conventional bulk carrier and containership orders. Meanwhile, POSCO's integrated steel complex at Pohang and Gwangyang remains one of the most operationally efficient steelmakers in the world, supplying the materials foundation for both shipbuilding and Korea's broader industrial economy.

Part I — Shipbuilding: The LNG Cycle and Beyond

The Three Korean Giants

Korean shipbuilding is concentrated in three vertically integrated conglomerates, each with facilities on the southeastern coast:

HD Korea Shipbuilding & Offshore Engineering (HD Group / formerly Hyundai Heavy)

- Facilities: Ulsan (Hyundai Heavy Industries main yard, largest single shipbuilding complex in the world at ~1.5km of berths), Gunsan (Hyundai Mipo Dockyard, mid-size vessels), Seosan (HSHI)
- Specialisation: LNG carriers, VLCCs, containerships, naval vessels
- 2025 revenue: KRW 29.93T; operating profit KRW 3.9T
- Backlog: Approximately 3.5 years of full production (end-2025 estimate)

Samsung Heavy Industries (Samsung Group)

- Facilities: Geoje Island (integrated shipbuilding complex)
- Specialisation: LNG carriers, offshore units (FPSOs), drillships, containerships
- 2025 net revenue: approximately KRW 10.5T; first full-year operating profit since 2015

Hanwha Ocean (Hanwha Group, formerly DSME)

- Facilities: Geoje Island (adjacent to Samsung Heavy)
- Specialisation: LNG carriers, submarines (KSS-III class), warships, LNG-FSRUs
- 2025 revenue: approximately KRW 8.2T; profitability restored after DSME's 2022 crisis

Together, the three Korean yards hold approximately 35% of global new shipbuilding orders by compensated gross tonnage (CGT) — but crucially, 70-75% of high-value orders (LNG carriers, offshore units, naval vessels) where the revenue-per-CGT ratio is 2-3× conventional vessels.

The LNG Carrier: Korea's Undisputed Domain

Korea's dominance in LNG carrier construction is the result of a 25-year technological investment that Chinese competitors have been unable to replicate at acceptable quality levels. An LNG carrier is one of the most technically complex commercial vessels ever built: it must contain cryogenic cargo (-163°C) in massive insulated tanks (Type A IMO prismatic or GTT Mark III membrane type), maintain hull integrity under continuous thermal cycling stress, propel 260,000-300,000 cubic meters of cargo at 19-21 knots, and operate continuously for 25 years with minimal downtime.

Korean yards dominate this market for several interconnected reasons:

1. GTT technology licensing: French firm GTT (Gaztransport & Technigaz) licenses its membrane LNG containment system exclusively to Korean (and one French/European) shipbuilders — explicitly excluding Chinese shipbuilders due to concerns about intellectual property protection. GTT's Mark III membrane system is used in approximately 80% of new LNG carriers globally.
2. Welding and metallurgical expertise: Invar steel (36% nickel alloy), used for GTT membrane systems, requires specialist welding that Korean yards have developed through decades of accumulated experience. Chinese yards that have attempted to build LNG carriers using alternative containment systems have had quality issues that eroded customer confidence.
3. Classification society relationships: DNV, Lloyd's Register, and Bureau Veritas have extensive track records with Korean LNG carriers; their surveyors have detailed knowledge of Korean yard practices. This institutional familiarity creates systematic quality assurance advantages.

The LNG carrier market is in a structural boom. Europe's post-Ukraine pivot from Russian pipeline gas to LNG imports requires approximately 150+ new LNG carriers over 2023-2030 to transport US, Middle Eastern, and Australian LNG to European terminals. Qatar's QatarEnergy massive LNG expansion (targeting 126 MTPA by 2027) placed the largest single shipbuilding order in history in 2021-2022: 100+ LNG carriers divided among Korean yards (HD HHI, Samsung Heavy, Hanwha Ocean). The economic value of this order portfolio to Korea is approximately \$35-40 billion over its delivery schedule.

The China Challenge: Real But Bounded

China's shipbuilding ambitions are well-documented. CSSC (China State Shipbuilding Corporation), Yangzijiang Shipbuilding, and other Chinese yards now account for approximately 55% of global new shipbuilding orders by CGT — more than Korea and Japan combined. In containerships and bulk carriers, Chinese price competition has been decisive: new Panamax containership prices from Chinese yards run approximately 10-15% below equivalent Korean quotes, reflecting lower labour costs, subsidised steel inputs, and state-backed financing for Chinese buyers.

However, the containership/bulk carrier market is not where Korean profitability resides. Korean yards have rationally ceded commodity vessel segments to China while concentrating capacity on LNG, specialty vessels, and offshore. The result is a bifurcated global market: China dominates volume, Korea dominates value. The question for the next decade is whether Chinese builders can climb the technology ladder into LNG carriers and complex offshore units. The GTT licensing barrier, the invar welding

expertise gap, and the regulatory/quality perception issues (Chinese-built complex vessels have had historical quality incidents that created reputational drag) suggest China's invasion of premium shipbuilding will be slow.

Offshore Energy: The Green Transition Opportunity

Korea's offshore engineering capability — accumulated through FPSO and drillship construction for major oil companies — is being repositioned for the offshore wind market. Hyundai Heavy's offshore engineering division has developed fixed-foundation and floating offshore wind platform concepts; Samsung Heavy has received engineering contracts from European offshore wind developers. The global offshore wind market, projected to require 500+ GW of new capacity by 2035, will demand hundreds of wind turbine installation vessels (WTIVs), offshore substations, and floating platforms — all product categories where Korean yards have applicable technology.

Part II — POSCO: The Steel Foundation

The Pohang-Gwangyang Duopoly

POSCO (now legally POSCO Holdings) operates two integrated steel complexes that together represent the most operationally efficient large-scale steel production in the world outside of China:

Pohang Works (North Gyeongsang, founded 1973): 17.5 million tonnes annual crude steel capacity; 10 blast furnaces; the birthplace of Korean industrial steelmaking under Park Chung-hee and founding POSCO CEO Park Tae-joon

Gwangyang Works (South Jeolla, founded 1987): 21 million tonnes annual crude steel capacity; 5 blast furnaces; the largest single-site steel complex in Korea and one of the top 5 globally

Together, POSCO produces approximately 38 million tonnes of crude steel annually. The product mix is premium: automotive steel (POSCO supplies Samsung SDI's battery materials production, Hyundai Motor, Kia), electrical steel (for EV motors and power transformers), stainless steel, and specialised shipbuilding steel grades. POSCO's Finex direct reduced iron process — a proprietary technology that uses iron ore fine powder directly (eliminating the coking coal pre-processing step) — provides a cost and environmental advantage over conventional blast furnace routes.

The 2022 Typhoon Hinnamnor Disaster and Recovery

POSCO's Pohang Works suffered catastrophic flooding in September 2022 when Typhoon Hinnamnor caused the Hyeongsan River to overtop its banks. The flood inundated POSCO's electrical systems, damaging approximately 30% of Pohang's production capacity and destroying equipment valued at approximately KRW 2.7 trillion (\$2B). The recovery — Pohang returned to full production within approximately 9 months through an extraordinary mobilisation of POSCO's internal engineering resources and contractor workforce — is a case study in industrial crisis management.

The Pohang recovery highlighted a longer-term vulnerability: the facility's location in a tidal river delta, combined with climate-change-driven typhoon intensification, represents an ongoing flood risk. POSCO has invested KRW 800+ billion in flood protection infrastructure post-2022, but the climate risk to Pohang's infrastructure is a legitimate long-term concern that appears in institutional ESG analysis of POSCO.

Battery Materials: POSCO's Strategic Pivot

POSCO's most important strategic decision of the 2020s is its transformation from pure steelmaker to integrated materials company. Through POSCO Future M (formerly POSCO Chemical), the group has built:

- Cathode Active Materials (CAM): NMC cathode production capacity for 90,000 tonnes annually by 2025, targeting 220,000 tonnes by 2030; primary supplier to LG Energy Solution and SK On
- Anode Materials: Natural graphite and synthetic graphite anode production at Sejong facility
- Lithium: POSCO Holdings has acquired lithium brine assets in Argentina (Sal de Oro) with 100,000 tonne LCE/year potential; the company is investing \$4B+ in lithium refining capacity in Argentina and Korea
- Nickel: POSCO International (trading subsidiary) operates nickel mining and smelting assets in Indonesia (Ramu NiCo) and Australia

The battery materials pivot transforms POSCO from a commodity steel producer into a vertically integrated battery supply chain player — with positions across mining, precursor materials, cathode active materials, and potentially cells. If Korea's battery industry maintains market position through the EV transition, POSCO's integrated materials platform will be one of Korea's most strategically valuable industrial assets.

Part III — Hyundai Steel and the Secondary Producers

Hyundai Steel: The Automotive Specialist

Hyundai Steel, Korea's second-largest steelmaker (approximately 23 million tonnes crude steel capacity), differs from POSCO in one critical dimension: it is captively integrated into the Hyundai Motor Group supply chain. Approximately 50-60% of Hyundai Steel's output goes to Hyundai Motor, Kia, and Hyundai Mobis — creating a structure where Hyundai Steel's fortunes are directly linked to Hyundai's automotive production volumes rather than market steel pricing.

This captive structure is double-edged. In an automotive downturn, Hyundai Steel's capacity utilisation drops alongside Hyundai Motor production; in a boom, Hyundai Steel benefits from premium pricing to a captive customer. The automotive steel market's transition to high-strength steels (AHSS — Advanced High Strength Steel) for EV lightweighting creates a product development opportunity: Hyundai Steel's ability to co-develop ultra-high-strength grades with Hyundai Motor's body engineers is a technical advantage versus commodity steel suppliers.

The China Overcapacity Shadow

All Korean steel producers operate under the shadow of Chinese steel overcapacity. China produced approximately 1.01 billion tonnes of crude steel in 2025 — more than half of global production — against domestic demand of approximately 870-900 million tonnes. The 100-150 million tonne surplus is priced at marginal cost and exported globally, with destination markets in Southeast Asia, India, and even Europe (before EU Carbon Border Adjustment Mechanism implementation). Korean domestic steel prices are structurally compressed by Chinese export competition; the average hot-rolled coil price differential between Korean domestic production and Chinese import arbitrage has been approximately \$50-80/tonne through 2024-2025.

POSCO's response has been product mix upgrade (moving up the value chain into automotive, electrical, and stainless steels where Chinese competition is less acute), geographic diversification (JV steel plants in India via POSCO Maharashtra Steel), and the battery materials pivot. These strategies are correct directionally but do not eliminate the cost-structure disadvantage versus Chinese integrated mills that operate with lower labour costs and subsidised energy.

Part IV — The Hydrogen Steel Transition

POSCO's HyREX Programme

POSCO has announced its HyREX (Hydrogen Reduction) programme, targeting the elimination of coal from steelmaking through direct reduced iron (DRI) using green hydrogen. The technical concept is well-established: DRI using hydrogen produces iron with water (H₂O) as the only byproduct, versus blast furnace routes that produce approximately 1.8-2.0 tonnes CO₂ per tonne of steel.

POSCO's timeline targets commercial-scale HyREX demonstration by 2030, with full transition pathway to net-zero steel by 2050. The challenges are formidable: green hydrogen production at the volumes required to replace coking coal in 38 million tonnes of annual steel production would require hundreds of gigawatts of electrolyser capacity and renewable electricity — infrastructure that does not exist at that scale today anywhere in the world.

The more immediately practical transition is DRI using natural gas (blue hydrogen from LNG with carbon capture), a pathway that POSCO is pursuing in partnership with Australian resource companies. POSCO's Saudi Arabia investment (considering a DRI plant using Saudi natural gas) would produce lower-carbon iron that can be shipped to Korea for electric arc furnace (EAF) steelmaking — bypassing the coal dependency of blast furnace routes.

Part V — Strategic Outlook

Shipbuilding's Next Decade: The Ammonia and Methanol Frontier

Korean shipbuilders' next competitive challenge is the development of ammonia-fuelled and methanol-dual-fuel vessels. The International Maritime Organisation's (IMO) 2023 revised GHG strategy — targeting net-zero by 2050 from shipping — means that LNG as a "transition fuel" will itself require replacement by 2040-2050. Ammonia (zero carbon combustion) and methanol (lower carbon than LNG) are the leading candidates for medium-term decarbonisation.

Korean yards are actively developing ammonia-ready engine configurations (Hyundai Heavy Industries has an ongoing programme with MAN Energy Solutions) and have secured early orders for methanol-fuelled containerships from Maersk and CMA CGM. The first ammonia carrier fleet (for green ammonia transport) is under construction at Korean yards for delivery 2027-2029. First-mover advantage in these new vessel types will be critical, and Korean yards are correctly positioning themselves as technology leaders in low-carbon marine propulsion.

POSCO's Materials Empire: The 2030 Aspiration

POSCO's ambition is to be the world's leading supplier of secondary battery materials by 2030 — supplying lithium, cathode materials, anode materials, and potentially separator components to the Korean battery triumvirate and to international battery producers. If Korea maintains its battery market position in the West (buttressed by IRA protections), POSCO's vertically integrated materials platform would be among the most strategically valuable industrial assets in Northeast Asia. The \$4B+ lithium investment in Argentina and the CAM expansion to 220,000 tonnes/year are the foundation of this aspiration. The risk is lithium price volatility (currently near multi-year lows) eroding the investment return during the ramp-up period.

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All data from company filings, Clarkson Research, POSCO Holdings, and cited news sources.